Bezawada Bennyhinn

sunnyashish41@gmail.com | +91 6300659698 | linkedin.com/in/benny-hinn-bezawada-58411331a | github.com/Benny45123

Summary

Software Engineer with hands-on experience building **voice-native AI systems**, including **real-time speech-to-text (STT)** and **text-to-speech (TTS)** pipelines, alongside production-grade backend infrastructure. Proven track record shipping end-to-end AI features: from audio ingestion and streaming inference to LLM-powered response generation and voice synthesis. Deeply comfortable with **real-time systems**, **audio ML**, and **agentic AI workflows** using LangGraph, LangChain, and vector databases.

Technical Skills

AI & LLMs: RAG, AI Agents, LangGraph, LangChain, LLM Orchestration, Prompt Engineering, Embeddings, Transformers, Diffusion Models, MCP, Pinecone, LiteLLM, STT, TTS, Whisper (Groq), Sarvam ai, Deepgram

Languages: Python, Java, JavaScript, TypeScript, C, HTML5, CSS3, SQL

Backend & Frameworks: FastAPI, Node.js, Express.js, Spring Boot, REST API Design, Microservices, Distributed Systems, WebSocket, Real-Time Systems, JUnit, Maven, Gradle

Databases: PostgreSQL, MongoDB, SQL, NoSQL, Redis, Vector Databases, Data Modeling

Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Git, GitHub Actions, Jenkins (CI/CD)

Core CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Projects

Remedi AI — Voice-Native Medical Triage Backend | *Python, FastAPI, WebSocket, STT, TTS, PostgreSQL* | [GitHub]

- Architected a **real-time voice AI backend** processing live patient symptom intake via **WebSocket streaming**, integrating **STT (Whisper, Deepgram)** for transcription and **TTS (Sarvam ai)** for AI-generated spoken triage responses, achieving sub-second end-to-end latency.
- Built a **multi-stage audio pipeline** with Voice Activity Detection (VAD) for intelligent chunking, noise filtering, and streaming inference, enabling seamless real-time conversation between patients and the AI triage agent.
- Designed REST and WebSocket APIs in **FastAPI** with async handling, PostgreSQL persistence for clinical logs, and structured LLM outputs for symptom-to-risk classification with strict guardrails and audit trailing.

Agentic AI Video Content Pipeline | *FastAPI, LangGraph, Supabase, MCP* | [GitHub]

- Architected a **multi-agent backend platform** using **LangGraph**, orchestrating Story, Script, Prompt, QA, Assembly, and Delivery agents to autonomously transform raw text into short-form videos.
- Integrated open-source **diffusion** and **transformer** video generation models (**Wan2.2 TI2V-5B**) self-hosted on **Kaggle GPU (T4)** infrastructure with external video-gen provider fallbacks.
- Implemented **Supabase (PostgreSQL)** checkpointing, conditional routing, Human-in-the-Loop approvals, and an **MCP tool layer**, reducing end-to-end generation latency by **75%+** via parallel execution.

Gov-Transparency-RAG | *Python, FastAPI, Pinecone, Redis* | [Frontend] | [Backend]

- Built a **full-stack RAG application** with a **FastAPI backend**, using **Gemini-embedding-001** and **Pinecone** to index **2,000+ federal court records** with BeautifulSoup and PyMuPDF data extraction.
- Designed a **Multi-Stage Retrieval Pipeline** combining semantic ANN search with Cohere reranking to deliver **high-precision factual retrieval** from sensitive legal documents while maintaining strict neutrality.
- Cut API costs and query latency by **40%+** via a **Redis caching layer**, asynchronous Supabase logging, and a custom batching pipeline engineered to handle high-throughput, rate-limit-constrained workloads.

Education

Vasireddy Venkatadri Institute of Technology — Sep 2023 – Present
B.Tech in CSE (IoT, Cybersecurity & Blockchain) | CGPA: 8.3 / 10 — *Guntur, India*

Achievements

Gold Medal – CodeVerse, Oct 2025 | Organized by Spaardha ACM (College-Level Tech Event) | [Link]

Cash Prize – Nexora Full-Stack Expo, Mar 2026 | Organized by IEEE | [Link]

AI Agents Certification – HuggingFace, 2026 | [Link]

AWS AI Practitioner Challenge – Udacity (AWS), 2026 | [Link]

500+ DSA Problems Solved – Java & Python | Competitive Programming & Interview Preparation